SET Lagrange points

Burden, rotation and the balance of field capacity

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11 October 2026

At a Lagrange point, the sources continue to burden the field. What balances is the motion of a probe that stays at one position relative to the rotating system. SET describes this through the same capacity relation used for the probe's momentum and inertia:

$$S^2 + |\mathbf{V}_{\mathrm{rel}}|^2 + c^2\beta^2 = c^2, \qquad \beta = \frac{d\tau}{dT} > 0. \tag{1}$$

Use Cartesian coordinates that rotate at constant angular velocity $\mathbf{\Omega}$. Let $\mathbf{r}$ be position in this chart and $\mathbf{v} = \mathrm{d}\mathbf{r}/\mathrm{d}T$ its coordinate velocity. The translation relative to the nonrotating frame, expressed in the rotating axes, is

$$\mathbf{w} = \mathbf{\Omega} \times \mathbf{r}, \qquad \mathbf{V}_{\mathrm{rel}} = \mathbf{v} + \mathbf{w}, \qquad \beta^2 = 1 - \frac{S^2 + |\mathbf{v} + \mathbf{w}|^2}{c^2}. \tag{2}$$

The spatial reading is the Cartesian one used at weak-field accuracy. The following exact identities refer to this specified capacity model. Define

$$U = \frac{S^2}{2}, \qquad F = U + \frac{|\mathbf{w}|^2}{2}, \qquad \beta_0^2 = 1 - \frac{2F}{c^2}. \tag{3}$$

Here β_0 is the clock rate of a probe held at fixed rotating coordinates, where $\mathbf{v} = 0$. It still translates with $\mathbf{w}$ in the nonrotating frame. Retain SET's probe action, with constant rest mass m:

$$L_p = -mc^2\beta, \qquad \mathbf{P} = \frac{\partial L_p}{\partial \mathbf{v}} = \frac{m(\mathbf{v} + \mathbf{w})}{\beta}. \tag{4}$$

For a field stationary in this rotating chart, a probe at fixed $\mathbf{r}$ has constant momentum components. The Euler–Lagrange equation therefore requires $\partial L_p/\partial \mathbf{r} = (m/\beta_0)\nabla F = 0$. Thus

$$\nabla F = 0 \quad \Longleftrightarrow \quad \nabla \beta_0^2 = 0. \tag{5}$$

This is the equilibrium condition derived from the capacity relation and the retained probe action. It identifies a stationary clock-capacity reading; the remaining burden need not vanish.

To express the directional balance, let $\mathbf{\Omega}$ lie along the z axis, and define the gravitational current reading by $\mathbf{J} = -\nabla U$. The rotating reading is

$$\mathbf{J}_{\text{rot}} = -\nabla F = \mathbf{J} - \Omega^2(x, y, 0). \tag{6}$$

At equilibrium, $\mathbf{J}_{\text{rot}} = 0$ and $\mathbf{J} = \Omega^2(x, y, 0)$. In the weak-field motion law, the acceleration is $-\mathbf{J}$, providing the centripetal acceleration needed to remain with the rotating system. **Directional balance leaves the local burden in place.** Neither $S = 0$ nor $\beta_0 = 1$ follows from it.

The same action gives the rotating energy. Taking its velocity transform yields

$$\mathcal{E} = \mathbf{P} \cdot \mathbf{v} - L_p = E - \mathbf{\Omega} \cdot (\mathbf{r} \times \mathbf{P}), \tag{7}$$

where $E = mc^2(1 - S^2/c^2)/\beta$ is the reference energy used in the inertia calculation. Equivalently,

$$\mathcal{E} = \frac{m}{\beta}\left(c^2\beta_0^2 - \mathbf{w} \cdot \mathbf{v}\right). \tag{8}$$

Since the rotating action has no explicit T dependence, $\mathcal{E}$ is conserved along its free-probe trajectories. The invariant also gives an exact algebraic gap:

$$\mathcal{E}^2 - m^2c^4\beta_0^2 = \frac{m^2}{\beta^2}\left[c^2\beta_0^2|\mathbf{v}|^2 + (\mathbf{w} \cdot \mathbf{v})^2\right] \geq 0. \tag{9}$$

For $\beta > 0$ and $\beta_0 > 0$, the rotating energy is positive. Equation (9) gives $\mathcal{E} \geq mc^2\beta_0$, with equality only at $\mathbf{v} = 0$. Define a conserved reading with speed-squared units:

$$C_{\text{cap}} \equiv c^2 - \frac{\mathcal{E}^2}{m^2c^2} = 2F - \frac{\beta_0^2|\mathbf{v}|^2 + (\mathbf{w} \cdot \mathbf{v})^2/c^2}{\beta^2}. \tag{10}$$

Consequently, $2F \geq C_{\text{cap}}$ is a necessary condition for a location to be accessible at the conserved energy. Equality is the zero-velocity boundary. In the joint weak-burden and low-speed limit,

$$C_{\text{cap}} \simeq 2U + |\mathbf{w}|^2 - |\mathbf{v}|^2. \tag{11}$$

This is the usual Jacobi combination for the circular binary model. The capacity expression connects equilibrium and allowed motion through the same energy reading.

The distinction is visible in the calculations below. In the Sun–Earth model, the gravitational current alone cancels about 259,000 km sunward of Earth; L_1 is about 1.492 million km away. For Earth–Moon, the corresponding distances are about 38,376 km and 58,019 km from the Moon. Rotation changes the balance required of the probe.

To calculate positions, specify the isotropic weak-field burden of two sources. Let d_i be the distance from the probe to source i. The source law rewrites its gravitational coefficient as

$$GM_i = \frac{Q_i^2}{24\pi V_i}, \qquad U = \sum_{i=1}^{2} \frac{GM_i}{d_i} = \sum_{i=1}^{2} \frac{Q_i^2}{24\pi V_i d_i}. \tag{12}$$

Here V_i is the reference volume of the individual source. This comparison uses the additive inverse-distance field and circular primary motion as stated weak-field inputs. Rewriting their coefficients in Q does not supply those inputs by itself. For separation a, set

$$\Omega^2 = \frac{G(M_1 + M_2)}{a^3}, \qquad \mu = \frac{M_2}{M_1 + M_2}. \tag{13}$$

Measure lengths in units of a and place the sources at (−μ, 0) and (1 − μ, 0). On the line through them, equation (5) becomes

$$x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} - \frac{\mu(x-1+\mu)}{|x-1+\mu|^3} = 0. \tag{14}$$

Its left-hand side increases strictly on each source-free interval, giving one collinear root in each: L_1 between the sources, L_2 beyond the second, and L_3 beyond the first. The off-axis conditions require both source distances to equal one and give

$$L_4, L_5: \qquad (x, y) = \left(\frac{1}{2} - \mu, \ \ \pm\frac{\sqrt{3}}{2}\right). \tag{15}$$

The table uses JPL DE440 gravitational parameters [1], with chosen circular separations of 1 au for Sun–Earth and 384,400 km for Earth–Moon. These are idealized model calculations, rather than dated ephemeris positions. The rotating current vanishes at all four points.

Circular model	Point	Distance to smaller source (10³ km)	S (km/s)	|J| (mm/s²)
Sun–Earth	L1	1492	42.340	5.871
Sun–Earth	L2	1502	41.918	5.990
Earth–Moon	L1	58.02	1.616	2.285
Earth–Moon	L2	64.51	1.388	3.156

[1] JPL Solar System Dynamics, Astrodynamic Parameters, DE440 values; accessed 11 October 2026. Adopted GM values in $m^3\ s^{-2}$: Sun, $1.32712440041279419 \times 10^{20}$; Earth, $3.98600435507 \times 10^{14}$; Moon, $4.902800118 \times 10^{12}$. One au = 149,597,870,700 m.

Equilibrium and stability ask different questions. Equation (5) locates a balance; stability asks how a nearby probe moves after a small displacement. The velocity response remains part of that calculation.

Let $\delta\mathbf{r}$ be a displacement from an equilibrium, let $\mathbf{w}_0$ and β_0 be evaluated there, and let $\mathbf{H}_F$ be the Hessian of F. Linearizing the same Euler–Lagrange equation gives

$$M_0\,\delta\ddot{\mathbf{r}} + 2\boldsymbol{\Omega}\times\delta\dot{\mathbf{r}} - H_F\,\delta\mathbf{r} = 0, \tag{16}$$

$$M_0 = I + \frac{\mathbf{w}_0\mathbf{w}_0^T}{c^2\beta_0^2}.$$

Dots denote derivatives with respect to T. The full inertial coefficient is $(m/\beta_0)\mathbf{M}_0$, the same velocity derivative obtained in *SET inertia*; the common factor cancels from equation (16). The rotational velocity term and the spatial derivatives of burden act together.

For the weak-field binary, $\mathbf{M}_0$ tends to the identity. At either triangular point, a planar perturbation proportional to $\exp(\lambda\Omega T)$ gives

$$\lambda^4 + \lambda^2 + \frac{27}{4}\mu(1-\mu) = 0. \tag{17}$$

After labeling the sources so that $0 < \mu \le 1/2$, the two planar mode pairs are oscillatory when

$$0 < \mu < \frac{1-\sqrt{23/27}}{2} \simeq 0.0385209. \tag{18}$$

The vertical mode is also oscillatory. The three collinear points each have a growing planar mode in this circular comparison. These reproduce the familiar weak-field stability conditions [2]; they describe linear response around the idealized equilibria.

A stationary capacity reading locates the balance. Its maximum or minimum alone cannot settle stability: the inertial and rotational velocity response remains essential.

SET describes a Lagrange point as a balance within an occupied field. Gravitational burden and translation both enter the clock capacity. Their spatial variations locate the equilibrium, and the same probe action determines momentum, conserved rotating energy and the response to displacement.

For the supplied isotropic binary field, this calculation recovers the standard weak-field positions and dynamics. Its exact capacity identities organize that result in SET variables; applying their higher-order terms to a real binary would also require its spatial response and source motion at the same accuracy.

[2] NASA Science, What are Lagrange Points?, for the classical equilibrium and stability comparison.